

19. All the compounds listed in **Column I** react with water. Match the result of the respective reactions with the appropriate options listed in **Column II**.

Column I

- A) $(\text{CH}_3)_2\text{SiCl}_2$
 B) XeF_4
 C) Cl_2
 D) VCl_5

Column II

- p) Hydrogen halide formation
 q) Redox reaction
 r) Reacts with glass
 s) Polymerization
 t) O_2 formation

ANSWER:
 A: p and s
 B: p and q and r and t
 C: p and q
 D: p

PART – II : MATHEMATICS

SECTION – I (Single Correct Choice Type)

20. For $r = 0, 1, \dots, 10$, let A_r , B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then

$$\sum_{r=1}^{10} A_r (B_{10}B_r - C_{10}A_r)$$

is equal to

- A) $B_{10} - C_{10}$ B) $A_{10} (B_{10}^2 - C_{10}A_{10})$ C) 0 D) $C_{10} - B_{10}$

ANSWER: D

21. Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to

- A) 25 B) 34 C) 42 D) 41

ANSWER: D

22. Let f be a real-valued function defined on the interval $(-1, 1)$ such that

$$e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt, \text{ for all } x \in (-1, 1), \text{ and let } f^{-1} \text{ be the inverse function of } f.$$

Then $(f^{-1})'(2)$ is equal to

- A) 1 B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) $\frac{1}{e}$

ANSWER: B

23. If the distance of the point $P(1, -2, 1)$ from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$, is 5, then the foot of the perpendicular from P to the plane is

- A) $\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$ B) $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$ C) $\left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3}\right)$ D) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2}\right)$

ANSWER: A

24. Two adjacent sides of a parallelogram ABCD are given by

$$\overrightarrow{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k} \text{ and } \overrightarrow{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$$

The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD' . If AD' makes a right angle with the side AB, then the cosine of the angle α is given by

- A) $\frac{8}{9}$ B) $\frac{\sqrt{17}}{9}$ C) $\frac{1}{9}$ D) $\frac{4\sqrt{5}}{9}$

ANSWER: B

25. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal was green is

- A) $\frac{3}{5}$ B) $\frac{6}{7}$ C) $\frac{20}{23}$ D) $\frac{9}{20}$

ANSWER: C

SECTION – II (Integer Type)

26. Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3}+1$ apart. If the chords subtend at the center, angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$, where $k > 0$, then the value of $[k]$ is
[Note : $[k]$ denotes the largest integer less than or equal to k]

ANSWER: 3

27. Consider a triangle ABC and let a , b and c denote the lengths of the sides opposite to vertices A, B and C respectively. Suppose $a = 6$, $b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to

ANSWER: 3

28. Let f be a function defined on $\mathbf{R}$ (the set of all real numbers) such that $f'(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4$, for all $x \in \mathbf{R}$.
If g is a function defined on $\mathbf{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in \mathbf{R}$,
then the number of points in $\mathbf{R}$ at which g has a local maximum is

ANSWER: 1

29. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying
 $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$.

If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to

ANSWER: 0

30. Let k be a positive real number and let

$$A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}.$$

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$, then $[k]$ is equal to

[**Note** : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k].

ANSWER: 4

SECTION – III (Paragraph Type)

Paragraph for questions 31 to 33.

Consider the polynomial

$$f(x) = 1 + 2x + 3x^2 + 4x^3.$$

Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$.

31. The real number s lies in the interval

- A) $\left(-\frac{1}{4}, 0\right)$ B) $\left(-11, -\frac{3}{4}\right)$ C) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$ D) $\left(0, \frac{1}{4}\right)$

ANSWER: C

32. The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval

- A) $\left(\frac{3}{4}, 3\right)$ B) $\left(\frac{21}{64}, \frac{11}{16}\right)$ C) $(9, 10)$ D) $\left(0, \frac{21}{64}\right)$

ANSWER: A

33. The function $f'(x)$ is

- A) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$
- B) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$
- C) increasing in $(-t, t)$
- D) decreasing in $(-t, t)$

ANSWER: B

Paragraph for Questions 34 to 36.

Tangents are drawn from the point $P(3, 4)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B.

34. The coordinates of A and B are

- A) $(3, 0)$ and $(0, 2)$
- B) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$
- C) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $(0, 2)$
- D) $(3, 0)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

ANSWER: D

35. The orthocenter of the triangle PAB is

- A) $\left(5, \frac{8}{7}\right)$
- B) $\left(\frac{7}{5}, \frac{25}{8}\right)$
- C) $\left(\frac{11}{5}, \frac{8}{5}\right)$
- D) $\left(\frac{8}{25}, \frac{7}{5}\right)$

ANSWER: C

36. The equation of the locus of the point whose distances from the point P and the line AB are equal, is
- A) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$
 B) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$
 C) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$
 D) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

ANSWER: A

SECTION – IV (Matrix Type)

37. Match the statements in **Column-I** with those in **Column-II**.

[**Note:** Here z takes values in the complex plane and $\text{Im } z$ and $\text{Re } z$ denote, respectively, the imaginary part and the real part of z .]

Column I

Column II

- | | |
|---|---|
| <p>A) The set of points z satisfying
 $z-i z = z+i z$
 is contained in or equal to</p> | <p>p) an ellipse with eccentricity $\frac{4}{5}$
 q) the set of points z satisfying $\text{Im } z = 0$</p> |
| <p>B) The set of points z satisfying
 $z+4 + z-4 = 10$
 is contained in or equal to</p> | <p>r) the set of points z satisfying $\text{Im } z \leq 1$</p> |
| <p>C) If $w = 2$, then the set of points
 $z = w - \frac{1}{w}$ is contained in or equal to</p> | <p>s) the set of points z satisfying $\text{Re } z \leq 2$</p> |
| <p>D) If $w = 1$, then the set of points
 $z = w + \frac{1}{w}$ is contained in or equal to</p> | <p>t) the set of points z satisfying $z \leq 3$</p> |

ANSWER:

- A: q and r**
B: p
C: p and s and t
D: q and r and s and t

38. Match the statements in **Column-I** with the values in **Column-II**.

Column I

Column II

A) A line from the origin meets the lines

p) - 4

$$\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1} \text{ and } \frac{x-\frac{8}{2}}{\frac{3}{2}} = \frac{y+3}{-1} = \frac{z-1}{1} \text{ at P and Q}$$

respectively. If length PQ = d, then d² is

B) The values of x satisfying

$$\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}\left(\frac{3}{5}\right) \text{ are}$$

q) 0

C) Non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$ satisfy $\vec{a} \cdot \vec{b} = 0$,

$$(\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0 \text{ and } 2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|.$$

If $\vec{a} = \mu\vec{b} + 4\vec{c}$, then the possible values of μ are

r) 4

D) Let f be the function on $[-\pi, \pi]$ given by

$$f(0) = 9 \text{ and } f(x) = \sin\left(\frac{9x}{2}\right) / \sin\left(\frac{x}{2}\right) \text{ for } x \neq 0.$$

s) 5

$$\text{The value of } \frac{2}{\pi} \int_{-\pi}^{\pi} f(x) dx \text{ is}$$

t) 6

ANSWER:

A: t

B: p and r

C: either q or (q and s)

D: r